"Oil"

Date
Page

☆ Optimization Problems:

→ A problem that may have many feasible solutions

→ Each solution has a value

→ In Maximization problem, we wish to find a solution to Maximize the value.

→ In Minimization problem, we wish to find a solution to Minimize the value.

☆ Few Techniques to solve such problems

- Greedy Algorithms
- Dynamic Programming
- Branch and Bound

☆ Greedy Algorithms

→ Many Optimization problems can be solved using a Greedy approach.

→ Basic principle is that Locally Optimal decisions may be used to build an Optimal solution

→ Greedy approach may not always lead

to an optimal solution overall for all problems.

→ Greedy algorithm always makes the choice that looks best at the moment.

→ The Hope: A locally optimal choice will lead to a globally optimal solution

→ Works for some problems.

→ Greedy algos are easier to code.

→ Greedy algos are simple.

→ Greedy algos are shortsighted in the sense that they take decisions on the basis of information at hand without worrying about the effect those decisions may have in the future.

→ Only few optimization problems can be solved by Greedy Method.

→ Greedy method solves the problem by making sequence of decisions.

→ Each decision is Locally optimal.

→ These Locally optimal solutions may add up to a globally

Optimal solution.

Example Making change

Coins of following Denominations are available

100
50
20
10
5
1

Problem: Make change of a given amount using the $\boxed{\text{Smallest}}$ possible number of coins

E.g. Make change of 283 Rs.

$283 > 100 \Rightarrow \frac{283}{100} = 2$ i.e. 2 coins X 100 Rs

283 Mod 100 = 83 → Remaining amount

$83 \not> 100$

$83 > 50 \Rightarrow \frac{83}{50} = 1$ i.e. 1 coin x 50 Rs

83 % 50 = 33 → Remaining amount

Date
Page

→ $33 \not> 100$
$33 \not> 50$
$33 > 20 \Rightarrow \frac{33}{20} = 1 \longrightarrow 1$ coin X 20 Rs

$33 \% 20 = 13$

→ $13 \not> 100$
$13 \not> 50$
$13 \not> 20$
$13 > 10 \Rightarrow \frac{13}{10} = 1 \longrightarrow 1$ coin X 10 Rs

$13 \% 10 = 3 \rightsquigarrow$ Remaining amount

→ $3 \not> 100$
$3 \not> 50$
$3 \not> 20$
$3 \not> 10$
$3 \not> 5$
$3 > 1 \Rightarrow \frac{3}{1} = 3 \longrightarrow 3$ coins X 1 Rs

$3 \% 1 = 0 \rightsquigarrow$ Remaining amount is 0

∴ Come out of algo

Sol: $2 \times 100 + 1 \times 50 + 1 \times 20 + 1 \times 10 + 3 \times 1$
Number of coins: $\boxed{8}$

Greedy Algo. for Making change

I/P: D[1...n], value

→ Array of Denominations

→ Assume D is in Descending Order of Denominations

E.g. D[1] = 100 D[2] = 50 D[3] = 20
D[4] = 10 D[5] = 5 D[6] = 2
D[7] = 1

O/P: Count of Coins

Algo:

```
Count = 0;  int ans[1...n] = {0};
for i = 1 to n
{
  ans[i] = Value / D[i];
  Value = Value % D[i];
  Count = Count + ans[i];
}
return Count and ans. arrays
```

O(n)

☆ This may not give optimal result always. ↓ why?

Ex: $D = \{1, 4, 6\} \longrightarrow \{6, 4, 1\}$ → 3 coins

Value = 8

Descending order of Denominations

I/P: $\{6, 4, 1\}, 8$

→ $8 > 6$ i.e. $\frac{8}{6} = 1$

$8 \% 6 = 2$ ⇝ remaining

→ $\frac{2}{4} = 0$

$2 \% 4 = 2$

$$\begin{array}{r} 0 \\ 4\overline{)2} \\ 0 \\ \hline 2 \end{array}$$

→ $\frac{2}{1} = 2$

$2 \% 2 = 0$

∴ 1 coin of 6 Rs

2 coins of 1 Rs

Total 3 coins but optimal answer is 2 coins of 4 Rs.